Advantages of payback method.

- Simplicity – simple to calculate & understood
- The method analyses cash flows not accounting profits. Investment are about investing cash to earn cash returns. Hence this method is better than ARR.

Disadvantages.

- It ignores all cash flows after the payback period and so ignores the total cash returns from the projects.
- It ignores the times of the cash flows during the payback period.

Example 2 Assessment

A company must choose between two investment, project A & project B. It cannot undertake both investments. The expected cash flows for each project are

Year	Project A	Project B
0	(80000)	(80,000)
1	20,000	60000
2	36000	24000
3	~~3~~ 30000	2000
4	17000	–

The company has a policy that the maximum permissible payback period for an investment is 3 years and if a choice has to be made, between two projects, the project with the earlier payback will be chosen.

Required.

1 – Calculate the payback period for each project.

Assuming:

a – that cash flows occur at that year end.

b – that cash flows after year 0 occur at a constant rate throughout each year.

2 – Which project ~~do~~ should be selected?

DISCOUNTED CASH-FLOW

Net present value (NPV) method of investment and the internal rate of return (IRR) method.

NPV methods

Steps involve.

1 – List all cash flows expected to arise from the project. (Initial investment, future cash inflows & future cash outflows).

2 – Discount these cash flows to their present values using the cost that the company has to pay for its capital (cost of capital) as a discount rate. The result is expresses in terms of todays value.

3. The NPV of the project is the difference between the present value of all the cost incurred and the present value of all the cash flow benefits. (savings or revenue)

Decision Rule:

- The project is accepted if the NPV is positive
- the project should be rejected if the NPV is negative

Example I.

A company with a cost of capital of 10% is considering investing in a project with the following cash flows.

Year	~~A~~(m)
0	(10,000)
1	6000
2	8000

Should the project be undertaken?

Exam 2

A company is considering whether to Invest in a new item of equipment costing ₦53,000 to make a new product. the product would have a four-year life, and the estimated cash profits over the four-year period are as follows:

Year	₦
1	17000
2	25,000
3	16000
4	12,000

Calculate the NPV of the Project using a discount rate of 11%

INTERNAL RATE OF RETURN (IRR)

Decision Rule:

A company might establish the minimum rate of return that it wants to earn on an Investment if ~~other~~ factors such as non-financial Considerations and risk and uncertainty are ignored.

* If the Project IRR is equal to or higher than the minimum acceptable rate of return, it should be undertaken. If otherwise, it should be rejected.

ESTIMATING THE IRR of an Investment.

To estimate the IRR, you should begin by calculating the NPV of the project at two different discount rate.

- One of the NPVs should be positive and the other NPV should be Negative

$$\text{Formula: } IRR = A\% + \left(\frac{NPV_A}{NPV_A - NPV_B}\right) \times (B - A)\%$$

Ideally the NPV at A% should be positive and the NPV of B% should be Negative.

Where

NPV_A = NPV at A%

NPV_B = NPV at B%

Example:

A Business requires a Minimum expected rate of return of 12% on its Investments. A proposed capital Investment has the following expected cash flows; at 10% & 15% respectively

Year	Cash flow
0	(80,000)
1	20,000
2	36,000
3	30,000
4	17,000
5	

Calculate the Internal rate of return of the project and advise whether to accept the project or reject it.

Example 2. The following information is about a project.

Year	Cash flows
0	(53,000)
1	17,000
2	25,000
3	16,000
4	12,000

This Project has an NPV of ~~#2,210 at~~ a discount rate of 11%

calculate the IRR at the discount rate of 5% and 8%. If the companys IRR is 10%, would you accept the project.